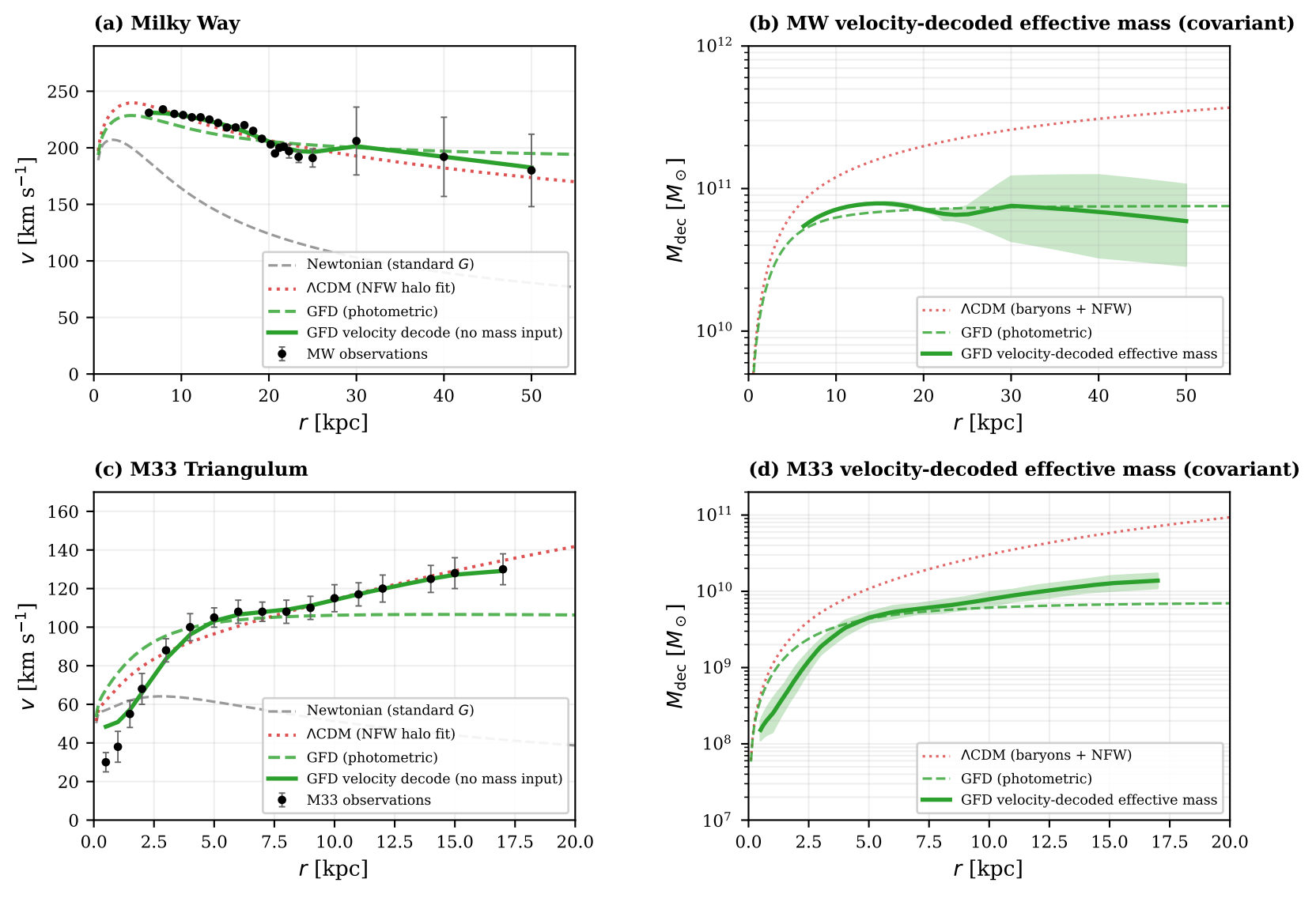

(a) Milky Way
v [km s⁻¹]
r [kpc]
Newtonian (standard G)
ΛCDM (NFW halo fit)
GFD (photometric)
GFD velocity decode (no mass input)
MW observations
(b) MW velocity-decoded effective mass (covariant)
M_dec [M_⊙]
r [kpc]
ΛCDM (baryons + NFW)
GFD (photometric)
GFD velocity-decoded effective mass
(c) M33 Triangulum
v [km s⁻¹]
r [kpc]
Newtonian (standard G)
ΛCDM (NFW halo fit)
GFD (photometric)
GFD velocity decode (no mass input)
M33 observations
(d) M33 velocity-decoded effective mass (covariant)
M_dec [M_⊙]
r [kpc]
ΛCDM (baryons + NFW)
GFD (photometric)
GFD velocity-decoded effective mass